



Department of Computer Science

**BSCCS Final Year Project Report
2022-2023**

22CS040

Portfolio Optimisation and Market Indexes Tracking

(Volume 1 of 1)

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Extended Abstract

This project aims to understand the significance of portfolio optimization for investors who seek to attain their financial objectives. Portfolio optimization refers to the process of choosing the optimal portfolio (asset allocation), from among all feasible portfolios, based on some criterion. The criterion usually involves maximizing factors such as expected return, and minimizing costs like financial risk. Portfolio optimization enables investors to manage the trade-off between risk and return, and to allocate their investments among various asset classes and securities. A prominent model for portfolio optimization is the Markowitz model, which was proposed by Harry Markowitz in 1952. The Markowitz model posits that investors are risk-averse and rational, and that they make their decisions based on the mean and variance of the portfolio returns. The Markowitz model assists investors in finding the efficient frontier, which is the set of portfolios that yield the highest return for a given level of risk, or the lowest risk for a given level of return. The efficient frontier indicates the optimal trade-off between risk and return for investors. However, the Markowitz model also has some drawbacks, such as neglecting higher moments of the return distribution, demanding a large number of inputs, and being prone to estimation errors. Hence, portfolio optimization continues to be a vibrant area of research and practice, with various methods and techniques being devised and applied to different asset classes and investment situations. This project will survey the literature on portfolio optimization, contrast different models and methods, and implement them to real-world data sets. This project aims to help evaluate certain formulations by applying them to various datasets of real-world stock prices on various indexes. Convex optimization has been used to help find the most optimal value for the evaluation of the different stocks on different ratios and risk measures.

Extended Abstract	i
1. Introduction	1
1.1. Problem Statement and Aim	1
1.2. Objective and Scope.....	1
1.3. Problem Significance and Extended Motivation	2
2. Literature Review	3
2.1. The Mean Variance Model	3
2.2. Continuous -time Model and Conditional-Value-at-Risk Optimizer	3
2.3. Naive Diversification	4
2.4. Sparse and Stable Optimisation	4
2.5. Theories.....	4
3. Proposed Design and Solution	10
4. Detailed Methodology and Implementation.....	12
4.1. Functions and Algorithms:	12
4.2. Experimental Setup:	14
5. Results and Future Improvement	16
6. Appendix.....	20
6.1. Bibliography.....	20
6.2. Monthly Logs	22

1. Introduction

1.1. Problem Statement and Aim

As previously noted in the Project plan, the optimization of portfolios using only the Markowitz method is mostly inefficient and unsatisfactory. The major purpose of this project is to optimize a variety of portfolio metrics, including the Sharpe ratio, Conditional Sharpe ratios, Sortino ratio, and a number of other risk measures, in order to enhance their performance. In 2007, DeMiguel, V. examined several portfolio construction algorithms affected by the Markowitz model. Given a sufficient amount of training data, the authors observed that none of the evaluated algorithms could consistently or significantly outperform the naïve strategy in which each available asset is allocated an equal weight. This unsatisfactory performance can be largely attributed to Markowitz's optimization strategy. Specifically, the optimization at the core of the Markowitz scheme is empirically unstable: small changes in expected asset returns, volatilities, or correlations can have significant effects on the output of the optimization technique. Standard Markowitz portfolio optimization is a weakly stated (or inadequately restricted) inverse problem in this regard. Numerous regularization strategies have been created to control the harmful instabilities that are frequently observed in numerous sectors.

1.2. Objective and Scope

The Project will emphasize the application of several risk measuring criteria, including the Sharpe ratio, Martin ratio, Pain ratio, and Sortino ratio. Combining the Sharpe Ratio formula with a weighted regularization parameter in order to balance the impacts of the Sharpe ratio and sparsity maximization. The project will then assess the performance of different portfolios based on these risk indicators, and it will be applied to numerous markets, including the S&P 500, FTSE100, HSCI, and HDAX. For instance, if the neurodynamic optimization (NO) portfolio is applied to the HSCI market, the outcomes of the NO portfolio will be compared to those of Equally distributed weighted optimization and others in terms of risk measures such as Sharpe Ratio, Sortino Ratio, to mention a few. Depending on the availability of official and high-quality data sets, more markets may be covered in order to estimate the optimization's success rate more precisely.

1.3. Problem Significance and Extended Motivation

The past has often demonstrated that value of assets may decrease unexpectedly and without warning. During the terrible times brought on by the COVID-19 pandemic, markets were significantly damaged at the commencement of the epidemic not too long ago. The stock market reached an all-time high on February 19, 2020, before the COVID-19 epidemic caused a steep collapse in share values. In the ensuing year, the world has changed, impacting our lives, our economy, and the fortunes of our enterprises; the shifting share prices reflect this journey in progress. The fundamental trends have increased, propelling some firms forward at unprecedented rates while others experience headwinds comparable to hurricanes. Diversifying the portfolio or improving it to get the desired market return is essential. Everyone has heard that portfolios should be diversified, but there is little agreement on what this involves. Unfortunately, the majority of investment portfolios are suboptimal; with minor modifications, they may have a higher predicted return, a lower risk profile, or both. This helped me to comprehend and become more interested in this issue, resulting in my selection of it for my final paper thesis.

2. Literature Review

2.1. The Mean Variance Model

Markowitz (1952) created the Mean-Variance (MV) model to solve this problem. In the model, the expected return is determined by the historical average of the stock's return, while the risk is determined by the standard deviation of these returns. The MV model treats the returns of individual assets as random variables and adopts the value of anticipated return and variance to measure return and investment risk, respectively (Zhang *et al.*, 2018).

If a buyer desires to reduce risk for a given fixed rate of return, the mathematical model for selecting stocks is (for a given level of risk, maximizing returns is the equivalent model):

$$\text{Minimize } Var[\xi_1 x_1 + \xi_2 x_2 + \dots + \xi_n x_n] \quad (1)$$

$$\text{s.t. } E[\xi_1 x_1 + \xi_2 x_2 + \dots + \xi_n x_n] \geq \alpha \quad (2)$$

$$x_1 + x_2 + x_3 + \dots + x_n = 1 \quad (3)$$

$$x_i \geq 0, i = 0, 1, 2, 3, \dots, n$$

If E is the expected value operation, Var is the variance operator, and x_i is the fraction of the total amount available for investment applied to stock i , ξ_i represents the random return for the i th stock (Zhang *et al.*, 2018). The main objective function for limiting risk is provided by Equation 1, in Equation 2 α represents the lowest acceptable rate of return for the investor, Equation 3 specifies that all available funds must be invested, while the non-negative constraint prohibits short sales.

2.2. Continuous -time Model and Conditional-Value-at-Risk Optimizer

Merton (1971) tackles the portfolio optimization problem by methodically constructing and analyzing optimum continuous-time dynamic models under uncertainty. According to the author, the primary benefit of the continuous-time model over discrete-time models is the fact that it considers just two types of stochastic processes: functions of Brownian motions and Poisson processes. Thus, the number of factors in the issue is lowered, allowing full utilization of the vast body of literature on these processes. According to Uryasev (2000), Conditional-Value-at-Risk (CVaR) is offered as a portfolio risk metric, hence optimization would be performed on top of this metric. In addition to the Value at Risk's cutoff point, the CVaR is generated from the weighted average of "severe" losses

in the tail of the distribution of probable returns (VaR). One of the benefits of CVaR is that it may be efficiently decreased with LP (Linear Programming) approaches.

2.3. Naive Diversification

DeMiguel et al. (2009a), and Behr et al. (2013) report that the uniform investment strategy (Naive diversification or 1/N) is rational when the degree of uncertainty about the risk/return distribution is high, and this fact is supported by a number of experimental studies demonstrating that this type of strategy tends to perform better in environments of extreme uncertainty, and is typically a good strategy for risk-averse investors. DeMiguel et al.(2009a) demonstrate when Naive Diversification is preferable to optimization, and Behr et al. (2013) confirm these findings by demonstrating that the optimization techniques are not uniformly superior to the 1/N strategy from the standpoint of estimating error outside the sample.

2.4. Sparse and Stable Optimisation

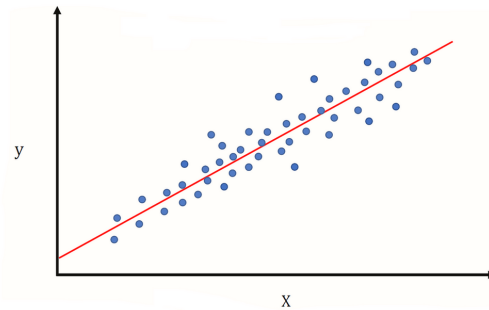
For mean-variance portfolio selection, sparse and stable optimization is a significant advancement. In general, DeMiguel et al. (2009b), Fan et al. (2012), Behr et al. (2013), and Dai & Wen (2018) advocate for the inclusion of weights that can take negative or positive values (Long and Short positions). The objective is to stabilize the asset weights in order to decrease estimate error using the Mean-Variance model with the limitation that the mean of the portfolio weight vector is smaller than a specific limit. The total weight may be less than 1 or equal to 1 (1-norm). In addition, the model admits negative values (Short), although normalization converts these to positive values. In addition, the threshold for the total of the weights must be larger than or equal to 1. This straightforward modification produced superior Sharpe outcomes compared to procedures described in the literature.

2.5. Theories

2.5.1. Linear Regression:

This approach for machine learning belongs to the supervised group. This algorithm predicts the values within a certain range rather than the category. For instance, it can forecast sales, pricing, etc. This method was utilized by researchers (Bhuriya, Kaushal, Sharma, & Singh, 2017) to forecast the

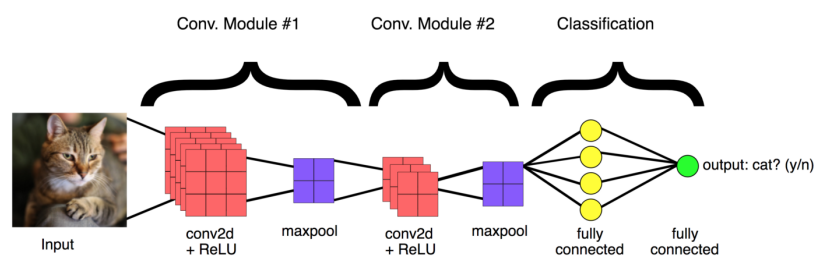
stock market. According to the experts, forecasting the daily stock value is a formidable obstacle that must be overcome. With this algorithm's predictions, investors may invest their funds with greater certainty.



Source: Stanford Online

2.5.2. Neural Networks:

This is the algorithm for unsupervised machine learning, which comprises of a number of nodes, each of which does some computation. Multiple academics have utilized this method to forecast stock prices. This method was utilized by researchers (Naeini, Taremian, & Hashemi, 2010) to forecast the stock market and help make better decision making for portfolio selection. Researchers (Adebiyi, Ayo, Adebiyi, Otokiti, & Sciences, 2012) built this algorithm with a variety of factors in order to anticipate the stock price so that investors may make the best selections. This algorithm may also be taught for this purpose using social media news feeds (Chen et al. (2017)).

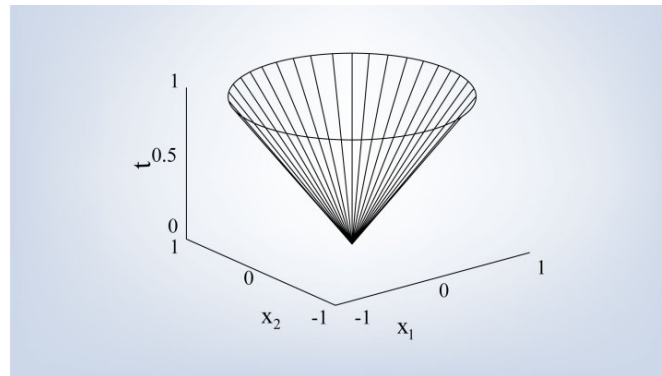


Source: TowardsDataScience

2.5.3. Convex Optimization:

A convex optimization problem is one in which the constraints are convex functions and the objective is a convex function if the aim is to minimize or a concave function if the objective is to

maximize. Since linear functions are convex, linear programming issues are also convex. The natural extension of linear programming problems, conic optimization issues are likewise convex.

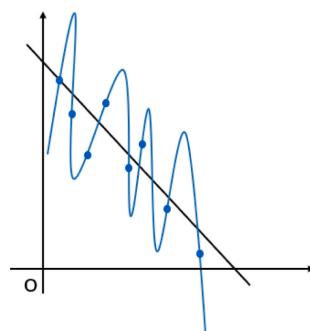


Source: Stanford Online

As seen above, the feasible zone of a convex optimization problem — the intersection of convex constraint functions — is a convex region. With a convex objective and a convex feasible region, there is only one globally optimum solution possible. Several approaches, most notably Interior Point methods, will either identify the globally optimal solution or demonstrate that no feasible solution exists. Convex problems can be effectively handled up to extremely large sizes.

2.5.4. Regularization:

The task of training a large-scale model is tough. A model may be chosen by maximizing its performance on a training data set. Its generalizability might be judged by its performance on unobserved data. In Fig. 1, for example, the data are fit to a linear function and a polynomial function. Despite the fact that the polynomial function matches the data more precisely, the linear function may be superior for predicting unknown data. The idea behind overfitting is that the loss lowers on training data, but rises on unseen data.



Source: Tian et al. (2021)

Due to the existence of input noise, the restricted size of the training set, and the complexity of classifiers, overfitting is a basic problem in supervised machine learning. Regularization is a crucial

aspect of machine learning , since it enables effective generalization to unknown data even when training on a limited training set or with insufficient iterations. Regularization involves the addition of non-obtained from the data restrictions to the minimized objective function, which indicate the previous preference.

2.5.5. Martin Ratio:

Peter G. Martin developed the Martin ratio as a measure of risk-adjusted performance for investments. The ratio of the investment's excess return over a risk-free rate to the ulcer index, which quantifies the magnitude and duration of drawdowns, constitutes the definition. Investment performance in relation to its downside risk is positively correlated with the Martin ratio, indicating that a higher Martin ratio is indicative of superior investment performance. The Martin ratio holds significance in the context of portfolio optimisation as it facilitates a comparative analysis of diverse investments or strategies by evaluating their risk-reward trade-off. The Ulcer Index is a metric that gauges the psychological ramifications of drawdowns on investors, and it imposes a greater penalty on significant and persistent losses compared to minor and brief ones. The Martin ratio is a useful tool for identifying the efficient frontier, which comprises portfolios that provide the maximum return for a given level of risk or the minimum risk for a given level of return. The efficient frontier is a representation of the most favourable balance between the level of risk and potential return that investors can achieve.

Martin Ratio Equation:

$$\begin{aligned}
 & \max_{y, \eta, \xi} \mu^T y - r_f \eta \\
 & \text{s.t. } e_n^T y = \eta, y \geq 0, \\
 & \frac{1}{m} \sum_{j=1}^m \left(\left(\xi_j - \sum_{i=1}^j r_i^T y \right)^+ \right)^2 \leq 1, \\
 & \xi_{j-1} \leq \xi_j, j = 1, 2, \dots, m;
 \end{aligned}$$

2.5.6. Calmar Ratio:

The Calmar ratio is a quantitative measure that evaluates the risk-adjusted returns of an investment vehicle, such as a hedge fund or a commodity trading advisor (CTA). The authorship of the work in question is attributed to Terry W. Young, and its initial publication occurred in 1991 within the periodical known as *Futures*, which is classified as a trade journal. The Calmar ratio is computed through the division of the mean annual compounded rate of return of a fund, typically spanning three years, by its maximum drawdown within the same duration. The maximum drawdown refers to the most significant decline in value experienced by a fund, measured from its highest point to its lowest point. A positive correlation exists between the Calmar ratio and the fund's risk-adjusted performance within the specified time period. The Calmar ratio holds significance in the context of portfolio optimisation as it enables investors to evaluate and contrast diverse funds or strategies by considering their risk-reward trade-off. The Calmar ratio is a risk assessment metric that solely employs the maximum drawdown as its criterion. This measure captures the psychological effects of losses on investors and imposes greater penalties on substantial and enduring drawdowns compared to minor and brief ones. The Calmar ratio is a useful tool for identifying the portfolio that provides the maximum return for a given level of risk or the minimum risk for a given level of return.

Calmar Ratio Equation:

$$\begin{aligned} & \max_{y, \eta, \xi} \mu^T y - r_f \eta \\ & \text{s.t. } e_n^T y = \eta, y \geq 0 \\ & 0 \leq \xi_j - \sum_{i=1}^j r_i^T y \leq 1, j = 1, 2, \dots, m; \\ & \xi_{j-1} \leq \xi_j, j = 1, 2, \dots, m \end{aligned}$$

2.5.7. Pain Ratio:

The pain ratio is also a quantitative measure that evaluates the risk-adjusted performance of an investment fund, including but not limited to hedge funds and commodity trading advisors (CTAs). The development of the aforementioned product was undertaken by Dr. Thomas Becker and Aaron Moore, both affiliated with Zephyr Associates. The calculation of the pain ratio involves the

division of the average annual compounded rate of return of a fund, typically spanning a three-year duration, by the fund's pain index over the same period, after subtracting the risk-free rate. The Pain Index is a quantitative metric that assesses the magnitude, duration, and frequency of drawdowns, which represent the maximum decline in the value of an investment portfolio. A positive correlation was observed between the level of pain ratio and the fund's risk-adjusted performance within the specified time period. The pain ratio holds significance in the optimisation of portfolios as it enables investors to evaluate and contrast various funds or strategies by taking into account their risk-reward trade-off. The pain ratio employs the pain index as a singular metric of risk, which takes into account the emotional ramifications of financial losses on investors and assigns greater weight to significant and sustained declines in asset value as opposed to minor and temporary ones. The employment of the pain ratio is a viable approach to determine the portfolio that is most advantageous in terms of providing the maximum return for a specific level of risk, or the minimum risk for a particular level of return.

Pain Ratio Equation:

$$\begin{aligned}
& \max_{y, \eta, \xi} \mu^T y - r_f \eta \\
& \text{s.t. } e_n^T y = \eta, y \geq 0, \\
& \quad \frac{1}{m} \sum_{j=1}^m \left(\xi_j - \sum_{i=1}^j r_i^T y \right)^+ \leq 1, \\
& \quad \xi_{j-1} \leq \xi_j, j = 1, 2, \dots, m;
\end{aligned}$$

Conditional Pain Ratio:

$$\begin{aligned}
& \max_{y, \rho, \eta, \xi} \mu^T y - r_f \eta \\
& \text{s.t. } e_n^T y = \eta, y \geq 0 \\
& \quad \rho + \frac{1}{m(1-\theta)} \sum_{j=1}^m \left(\xi_j - \sum_{i=1}^j r_i^T y - \rho \right)^+ \leq 1, \\
& \quad \xi_{j-1} \leq \xi_j, j = 1, 2, \dots, m.
\end{aligned}$$

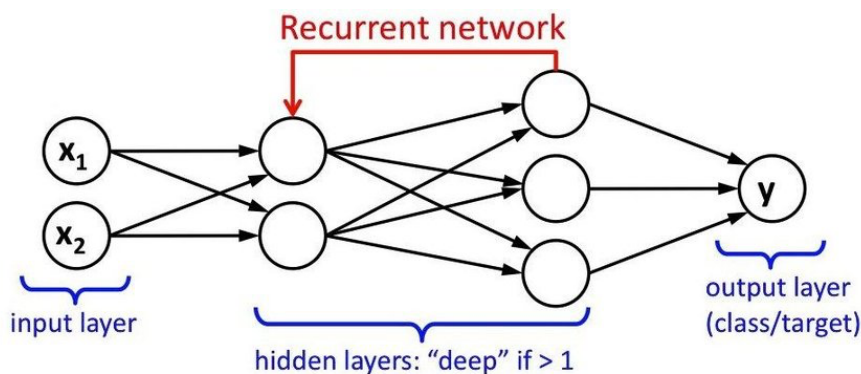
3. Proposed Design and Solution

The most significant technical aspect of the project is the usage of Matlab and other data visualization tools to analyze the optimization findings. It will be used to assist with the coding and interpretation of datasets and outcomes.

The datasets are gathered and cleaned to ensure the data's quality. The datasets are given by Professor Wang and his former team, who worked on the project before. Eventually, more fresh datasets will be discovered and evaluated to see whether they are more trustworthy. The Datasets would be used from different markets such as HSI, DAX100, FTSE100, NASDAQ 100, S&P100 to name a few. These datasets would be used for the verification of the reformulations of convex equation based ratio formulas. They will be tested for the maximizing of the Calmer ratio, the Pain ratio, the Martin ratio, and the Conditional Pain ratio with the family of Drawdown risk measures.

Applications in machine learning, neural networks, deep learning, computer vision, and autonomous driving may all make use of the MATLAB® software's particular toolboxes. With only a few lines of code, MATLAB makes it feasible to construct neural network models, generate and display them, connect them with existing applications, and deploy them to servers, corporate systems, clusters, clouds, and embedded devices. Their commercial and academic use serves will be used for the optimization problem and plotting of huge datasets results after the analysis.

The algorithm will be based on the problem reformulations based upon the research done by Professor Wang and his team in the past. The algorithm will be inclusive of the cardinality constraint when there is optimization with sparsity nature of the portfolio model using RNN



Recurrent Neural Networks Model

Convex optimisation refers to the procedure of minimising a convex objective function while being subject to convex constraints. Alternatively, it involves maximising a concave objective function while being subject to convex constraints. The MATLAB software package provides a pre-existing function, namely `fmincon`, which is capable of resolving convex optimisation problems through the utilisation of diverse algorithms, including interior-point, trust-region-reflective, active-set, or `sqp`. In order to utilise `fmincon`, it is necessary to explicitly define the objective function, initial point, linear and nonlinear constraints, lower and upper bounds, and solver options. Upon execution, `fmincon` will provide the optimal solution, optimal value, exit flag, and additional output information.

4. Detailed Methodology and Implementation

4.1. Functions and Algorithms:

4.1.1. Functions:

Fmincon:

The `fmincon` function in MATLAB is capable of resolving nonlinear constrained optimisation problems, which encompasses convex problems. Multiple algorithms, including interior-point, trust-region-reflective, active-set, or `sqp`, are utilised. In order to utilise `fmincon`, it is necessary to define the objective function, initial point, linear and nonlinear constraints, lower and upper bounds, and solver options. The `fmincon` function is capable of identifying a local minimum of a convex problem that is equivalent to a global minimum. The variability of outcomes is contingent upon the starting point and the level of continuity exhibited by the objective function and constraints.

PatternSearch:

The optimisation solver, `patternsearch`, implemented in MATLAB, is utilised to determine the minimum value of a constrained nonlinear multivariable function. This technique is employed to address optimisation problems that do not exhibit linearity or quadratic characteristics. The employed methodology of the function involves the utilisation of a derivative-free optimisation algorithm, specifically pattern search, to determine the minimum value of the objective function. The pattern search technique is a non-gradient-based direct search method utilised for optimising objective functions. Rather than employing a direct approach, the algorithm undertakes a search for the minimum value through a process of traversing the search space from one point to another.

GlobalSearch:

The MATLAB `GlobalSearch` function is a computational tool designed to solve optimisation problems by identifying the global minimum of a multivariable function subject to constraints. This method is utilised to address optimisation problems that do not exhibit linearity or quadratic characteristics. The employed optimisation algorithm in the function is `fmincon`, which is based on derivatives, and is utilised to determine the minimum value of the objective function. The `GlobalSearch` algorithm employs a scatter-search approach to produce initial points. The score function of a collection of `NumStageOnePoints` experimental points is assessed. Subsequently, the algorithm selects the point with the highest score and executes `fmincon` starting from said point.

4.1.2. Algorithms:

Sequential quadratic programming:

Sequential Quadratic Programming (SQP) is a mathematical optimisation technique that involves solving a series of quadratic subproblems to provide an approximation of the initial nonlinear problem. Every individual subproblem entails a quadratic estimation of the objective function and a linear estimation of the constraints based on the present iteration. The iterative process is updated through the resolution of a subproblem, followed by the application of either a line search or a trust region method. The algorithm additionally upholds a set of constraints that remain fulfilled as equalities at the present iteration. The cessation of the algorithm occurs upon the discovery of a point that fulfil the conditions of first-order optimality.

$$x = fmincon(fun,x0,A,b,Aeq,beq,lb,ub,nonlcon,options)$$

Where,

fun is the nonlinear objective function to be minimized.

x0 is the initial point for the algorithm.

A is the linear inequality constraint matrix.

b is the linear inequality vector.

Aeq is the linear equality constraint matrix.

beq is the linear equality vector.

lb is the lower bound for the variables.

ub is the upper bound for the variables.

nonlcon is a function that returns the nonlinear inequality and equality constraints, respectively.

options is a structure that specifies additional options for the algorithm, such as 'sqp' for the Algorithm option.

Active Set:

The active-set approach is a computational technique that addresses a series of quadratic subproblems while maintaining a constant active set of constraints during each iteration. The algorithm commences by initialising an active set, followed by solving the associated quadratic subproblem. Subsequently, the algorithm verifies if the present iteration fulfils the first-order optimality prerequisites. In the event that the specified conditions are not met, the algorithm proceeds to update the active set through the addition or removal of constraints, and iterates this

process until convergence is achieved. The algorithm is capable of addressing certain issues related to nonsmooth constraints, albeit its performance may be suboptimal when dealing with problems of significant scale.

Interior Point:

The interior-point approach involves the conversion of the initial problem into an equivalent form that solely comprises inequality constraints, accomplished through the incorporation of slack variables. The method in question employs a barrier approach to address a sequence of barrier subproblems. Each subproblem introduces a logarithmic barrier term into the objective function, thereby preventing the iterates from contravening the inequality constraints. The resolution of each barrier subproblem is accomplished through the utilisation of a primal-dual Newton method. This method involves the computation of the search direction via the solution of a system of linear equations, specifically the Karush-Kuhn-Tucker (KKT) system, at every iteration. As the iterations progress, the algorithm gradually decreases the barrier parameter and converges towards the initial problem. The algorithm is capable of efficiently managing problems that are both large-scale and sparse. However, it necessitates strict feasibility of the problem, meaning that there must exist a point that satisfies all the inequality constraints in a strict manner.

4.2. Experimental Setup:

Yahoo Finance was the source for the acquisition of these datasets. The datasets include a smaller number of equities compared to the number of index components that are currently traded in the marketplaces. The occurrence that was found may be explained by the fact that some companies were omitted from the datasets in the event that their admission or deletion from the indexes occurred within the time frame of the investment horizon. Beasley's OR-Library (Beasley, 1990), Bruni-Cesarone-Scozzari-Tardella-Library (BCST-Library) (Bruni et al., 2016), and four other datasets that were previously updated and utilised in research that was conducted by Leung and Wang (2021, 2022), and Leung et al. (2022a,b) are all incorporated into the experimental design of the study.

The datasets have been partitioned into two equal halves based on the number of weeks that pass between each observation. The remaining 50% of the dataset is put aside for out-of-sample testing, especially portfolio rebalancing, while the first 50% is set aside for in-sample learning of issue

parameters such as μ and V . Utilising the first fifty percent of the data that is available is the first step in the process of in-sample pre-training. The next step is to update the information with the most current pricing data just before the portfolio is rebalanced. The BCST-Library contains a number of datasets, each of which has an odd number of weeks. These datasets are organised in such a way that they leave out the data for the last week. This is done in order to facilitate the development of partitions that are split in a consistent manner.

These curated datasets provide information on the prices or returns of key global stock markets and indexes that have been modified on a weekly basis. The above stated stock indexes are traded on a variety of stock exchanges located all over the globe. A successor to the DAX100 (HDAX), the Deutsche Borse (DAX100) is now one of the indices that may be traded on the Deutsche Borse. The Stock Exchange of Hong Kong is where transactions for the Hang Seng Composite LargeCap and MidCap Index (HSCI) take place. The Dow Jones Industrial Average (DJIA), Standard & Poor's 500 (S&P500), and NASDAQ100 are traded on the New York Stock Exchange and the NASDAQ System, respectively. The Financial Times (FTSE100) is listed on the Stock Exchange of London. These databases have recently undergone modifications, and as a result, they now include the revised closing prices of a significant fraction of the equities that are traded on the four most important stock exchanges throughout the globe.

Table 1 - Dataset and Libraries Description

Library	Dataset	No. of Stocks	No. of Weeks	Time Span	Reference
BCST-Library	S&P500	442	595	01/11/2004 - 29/04/2016	(Bruni et al., 2016)
	NASDAQ100	82	596	01/11/2004 - 29/04/2016	
	FTSE100	83	717	01/07/2002 - 29/04/2016	
	DJIA	28	1363	01/02/1990 - 29/04/2016	
Updated Library	HDAX	49	938	03/01/2000 - 29/11/2017	(Leung et al., 2022a)
	FTSE100	56	938	03/01/2000 - 29/11/2017	
	HSCI	77	938	03/01/2000 - 29/11/2017	
	S&P500	356	938	03/01/2000 - 29/11/2017	

5. Results and Future Improvement

5.1. Testing Results:

After running the code and verifying with some of the pre-tested results, new codes were done for Martin Ratio, Pain Ratio, Calmar Ratio, and Conditional Pain Ratio. The codes were inspired by some of the previous works done by Professor Wang and his team for the Neural Networks Paper (Wang et al., 2023). Some of the test results mentioned in Table 2 and 3 are the implementation of Calmar Ratio, and Pain Ratio on the Updated Library (Leung et al., 2022a). Followed by implementation of Calmar Ratio and Martin Ratio on BCST-Library (Bruni et al., 2016). The Updated library compares the values between the neurodynamic optimization (NO) along with equality weighted (EW) portfolios and market indices (MI) portfolios. The BCST-Library compares values between neurodynamic optimization (NO) along with equality weighted (EW), market indices (MI), the Cumulative Zero- order epsilon Stochastic Dominance (CZeSD), Markowitz's mean-variance (M-V), Lizyayev-Ruszczynski's approximate second-order stochastic dominance (LR-ASSD), Post- Kopa's Second-order stochastic dominance method (PK- SSD), Roman-Mitra-Zviarovich's second-order stochastic dominance (RMZ-SSD), Luedtke's second-order stochastic dominance (L-SSD) portfolios.

It was evident that the neurodynamically optimised portfolios performed better in most cases not just for the ratios mentioned below but as well for the previously tested ratios such as Sharpe, Sortino, and Omega ratios to name a few. For the updated libraries the NO portfolios performed better than the baselines in about ten out of fifteen sets across four ratios and performed second best in five of these fifteen. For the BCST libraries across the three ratios which comprises of twelve different cases and comparing with eight other baselines, it performed best (including second best) in seven cases.

Table 2- Updated Library: The ex-post ratios of the neurodynamically optimised portfolios compared with equally weighted and market indices portfolios, where the best result is bolded and the second best is underlined.

Ratios	Dataset	Portfolios		
		NO	EW	MI
Calmar Ratio	HDAX	0.0766	<u>0.0649</u>	0.0549
	FTSE100	0.1029	<u>0.0927</u>	0.0427
	HSCI	0.0978	<u>0.0709</u>	0.0314
	S&P500	<u>0.0827</u>	0.0994	0.0691
Martin Ratio	HDAX	0.2721	<u>0.2408</u>	0.2073
	FTSE100	0.3706	<u>0.3441</u>	0.1497
	HSCI	<u>0.1873</u>	0.2278	0.0936
	S&P500	0.0255	0.0187	<u>0.0232</u>
Pain Ratio	HDAX	0.4288	<u>0.3764</u>	0.3240
	FTSE100	0.5585	0.5189	0.2228
	HSCI	<u>0.2683</u>	0.3335	0.1264
	S&P500	<u>1.384</u>	1.426	1.298
Conditional Pain Ratio	HDAX	0.0975	<u>0.0865</u>	0.0793
	FTSE100	0.1311	0.0985	<u>0.1196</u>
	HSCI	<u>0.0855</u>	0.0924	0.0783

Table 3 - BCST-Library: The ex-post ratios of the neurodynamically optimised portfolios compared with seven baselines, where the best result is bolded and the second best is underlined.

Ratios	Dataset	Portfolios								
		NO	EW	MI	M-V	L-SSD	LR-ASSD	RMZ_SSD	PK_SSD	CZeSD
Calmar Ratio	DJIA	0.0154	<u>0.0248</u>	0.0132	0.01722	0.0134	0.0195	0.0088	0.0284	0.0215
	NASDAQ 100	0.1939	0.1517	0.1391	0.1190	<u>0.1542</u>	0.1269	0.1075	0.1077	0.1260
	FTSE100	0.0843	0.01230	0.0434	0.2104	0.1660	0.0440	<u>0.1918</u>	0.1904	0.0918
	S&P500	0.1297	0.0986	0.0910	0.0947	0.1017	0.0968	0.1408	0.0784	<u>0.1300</u>
Martin Ratio	DJIA	<u>1.3900</u>	1.4373	0.4183	0.7935	0.3953	0.8723	0.1310	0.5979	1.2003
	NASDAQ 100	<u>1.3894</u>	1.5476	1.1642	1.3120	1.8891	1.0654	0.8113	0.4647	1.1492
	FTSE100	1.1374	0.8420	0.2126	2.7423	1.6577	0.2740	<u>2.2513</u>	0.7786	0.5102
	S&P500	<u>0.0947</u>	0.0901	0.0933	0.0832	0.0801	0.0730	0.0897	0.0301	0.1185
Pain Ratio	DJIA	<u>0.2934</u>	0.3078	0.1135	0.1567	0.1063	0.2017	0.0375	0.1669	0.2309
	NASDAQ 100	<u>1.1956</u>	1.1576	0.8020	0.8246	1.2699	0.8440	0.8075	0.7979	0.9103
	FTSE100	0.7921	0.6494	0.1991	<u>1.2198</u>	1.0319	0.4166	1.2473	0.8973	0.4614
	S&P500	<u>0.7907</u>	0.7572	0.6733	0.5331	0.5016	0.5733	0.6292	0.5023	0.9491

5.2. Further Improvements and Reflections:

The tests have been completed on the required tables however, there is chances of finding a more efficient way to solve the problem if the problems are reformulated in future. Furthermore testing on the Conditional pain ratio can be done for the BCST library. There can also be scope of working on the annualised return of these neurodynamically optimised portfolios.

I was able to learn a lot about the area I hadn't explored in the past, the region of MATLAB. I learnt quite a lot regarding what is convex optimisation, I also learnt about financial engineering and how it effects day to day activities such as trading. The use of advanced technologies along with some financial engineering can help one better understand and analyze the market. I would like to thank Professor Jun WANG and his team for helping me and guiding me through different situations to complete the project.

6. Appendix

6.1. Bibliography

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6.2. Monthly Logs

6.2.1. October Month:

Had two biweekly meetings with the professor. There was the project plan submission and discussion over the topics of which data sets to use for the project. I also researched the drawbacks and advantages of Neural networks, MATLAB, and Convex Optimization and read some papers based on the topics provided by Prof. Wang. The discussion with Professor Wang was fascinating as I got a different point of view on the topic and his latest amendment to his paper again provided valuable insights. I later set up the MATLAB environment and searched over the effects of RNN in python. This month I had a full-time job from Monday to Thursday, along with midterms, so I did not get enough time to meet with the professor and gain more insights. However, in the coming few weeks, as my priority is submitting the interim report, I will have more developments in the project work.

6.2.2. November Month:

Meeting with the professor in due course via online and physical modes. Discussing about the new reformulations and the set up of the experiment for testing purpose. There was progress on the literature reviews, not just based on the papers given by Professor Wang, but also finding out more papers by externals in the field of optimization, neural networks, and portfolio management. I learnt more about the RNN in the MATLAB environment. Observed more about the convex optimization server and other commercial servers. As this month had my midterms and other project deadlines with placement, I spend most of the time on learning. The last week of the month will be the beginning of the testing phase starting with the warm-up testing methods. Also, there is plan on having some discussion session with Prof. Wang's previous master students who also worked on the reformulations.

6.2.3. December Month:

There was submission of the Interim report 1. As it was exam season and post exam season I had only two meetings with the professor regarding future plans and experimenting timelines. Started work with data and MATLAB.

6.2.4. January Month:

Met the Professor regarding the new reformulated formulas. Got the new updated datasets along with some old reference codes for understanding the scheme and process of coding. Discussed some plans about experimentations and processes.

6.2.5. February Month:

Got connected with a PhD student help understand the paper and the formulations. As there was no direct student working on the project it took longer time to understand the paper and formulations. Learnt about MATLAB optimising functions. Started to write some pseudo code to check the working of the reference codes.

6.2.6. March Month:

Worked on Data sets of OR library and updated Library. Worked on the coding reformulation of the Sharpe, Sortanio, and Calmar. Drew outcomes on the BCST library for Calmar Ratio. Worked on the code for Martin and Pain Ratio after which will run on datasetsSpoke to PhD students to verify the coding method and realise the efficiency of the codes.

6.2.7. April Month:

Running on the code for Martin and Pain Ratio datasets. Guidance from PhD student on how to better optimize the code. Worked on the coding reformulation of the Pain, and Calmar. Drew outcomes on the BCST library for Calmar Ratio. As it was exam season could not focus much on the experimentation as I had both internship and school exams. Still made moderate progress in terms of experimentation.

6.2.8. May Month:

Trying to optimize the code for data output for Pain, Conditional Pain and Martin Ratio. Focusing on the preparation for the Interim Report 2. Running and writing codes for untested ratios. Completed tests on certain ratios for the two different libraries.

6.2.9. June Month:

Trying to optimize the code for data output for Pain, Conditional Pain and Martin Ratio. Focusing on the preparation for the Final Report. Running and writing codes for untested ratios. Completed tests on all ratios except Conditional Pain ratio for both libraries.

6.2.10. July Month:

Trying to optimize the code for data output for Pain, Conditional Pain and Martin Ratio. Focusing on the preparation for the Final report, codes, and presentation. Running and writing codes for untested ratios. Completed tests on all ratios for Updated library.